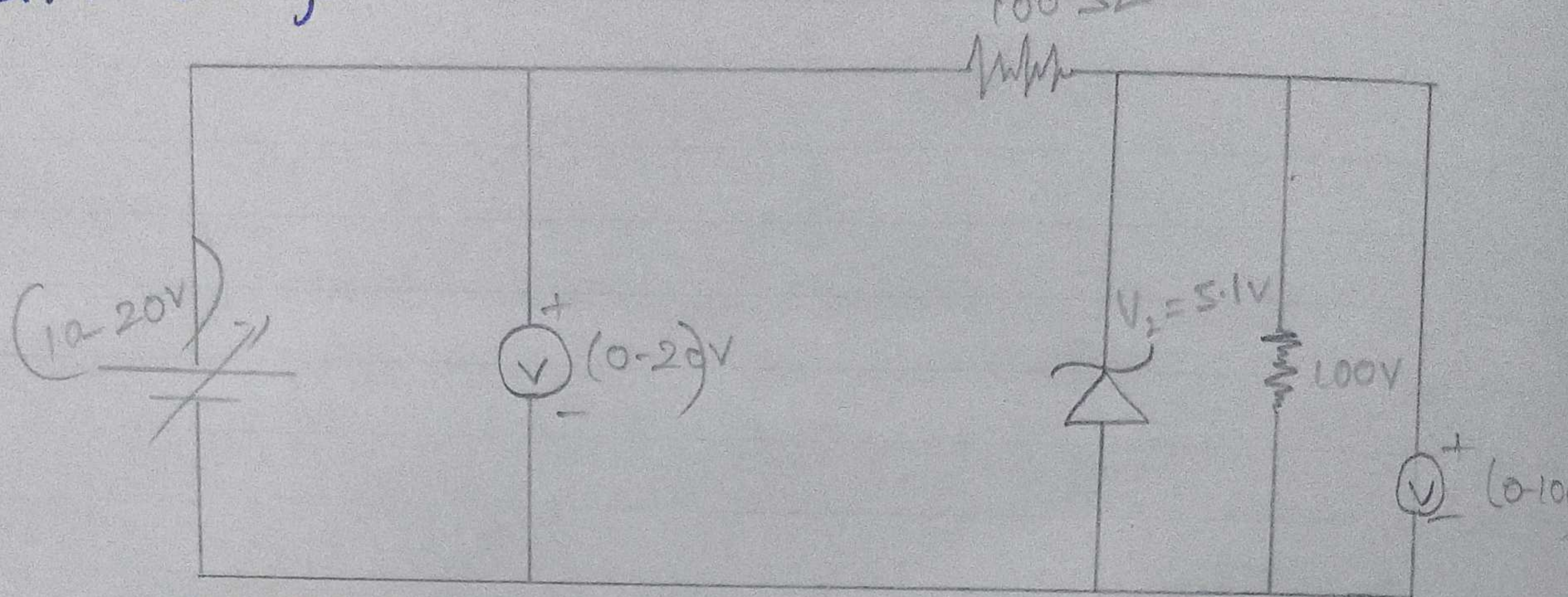
Line Regulation:

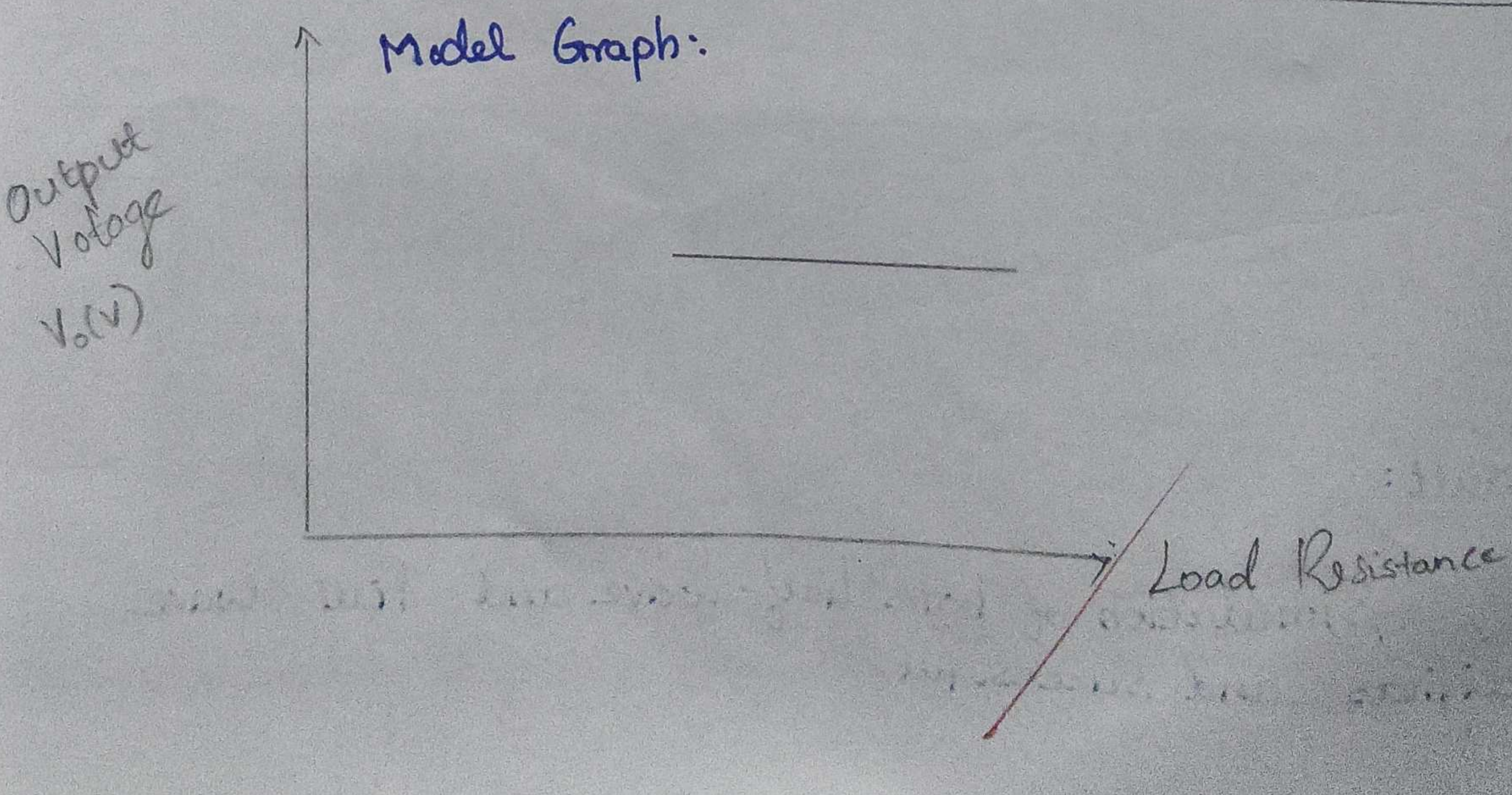
Circuit Diagram:



Tabulation:

Sl No.	Input voltage V_i (V)	Output voltage (V)
1	+12.0	+5.14
2	+13.0	+5.16
3	+14.0	+5.17
4	+15.0	+5.17
5	+16.0	+5.18
6	+17.0	+5.19

Model Graph:



09/3/26

Zener diode as voltage Regulator.

Aim:

To simulate the line and load regulation operation zener diode.

Apparatus Required:

Laptop with proteus Software.

Theory:

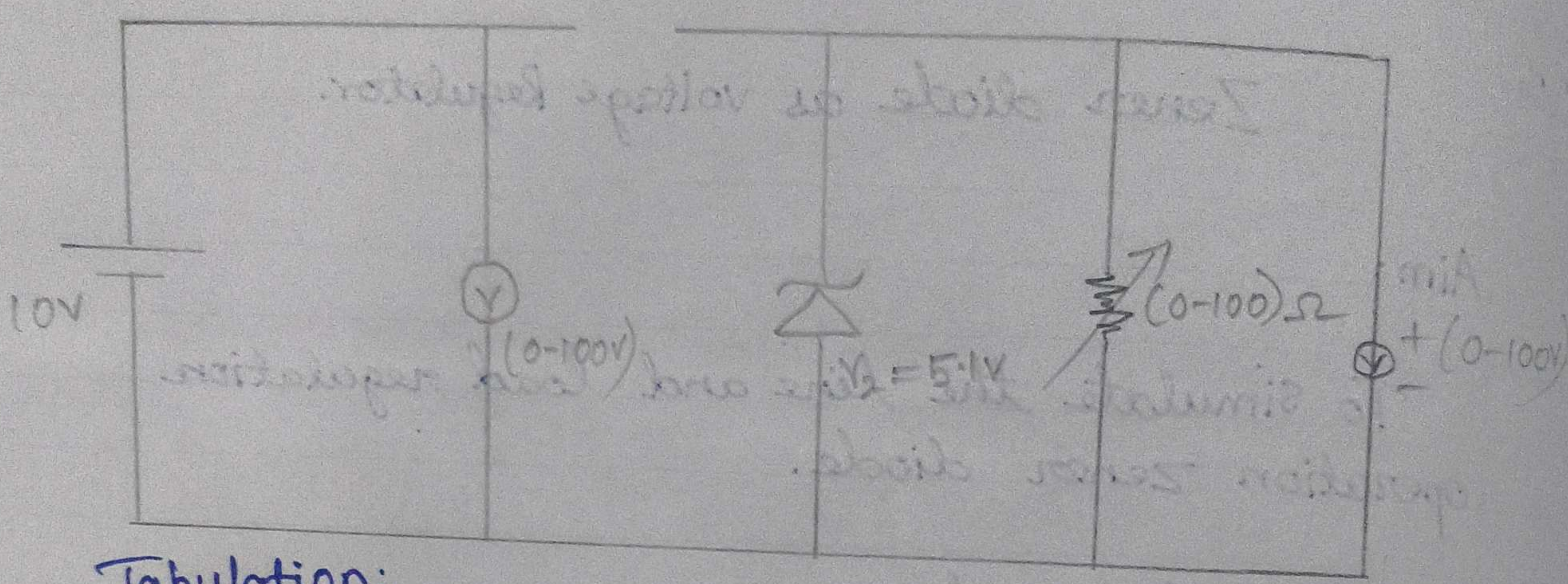
A heavily dropped PN Junction diode is called zener diode. Due to heavily dopped nature zener diode works under forward bias and reuses bias conditions.

Zener diode under reverse bias condition is used as a voltage regulator Even it input Voltage Changes the output is constant. This is called line regulation.

If the load resistance changes the output voltage is constant. This is called Load Regulation.

Load Regulation:

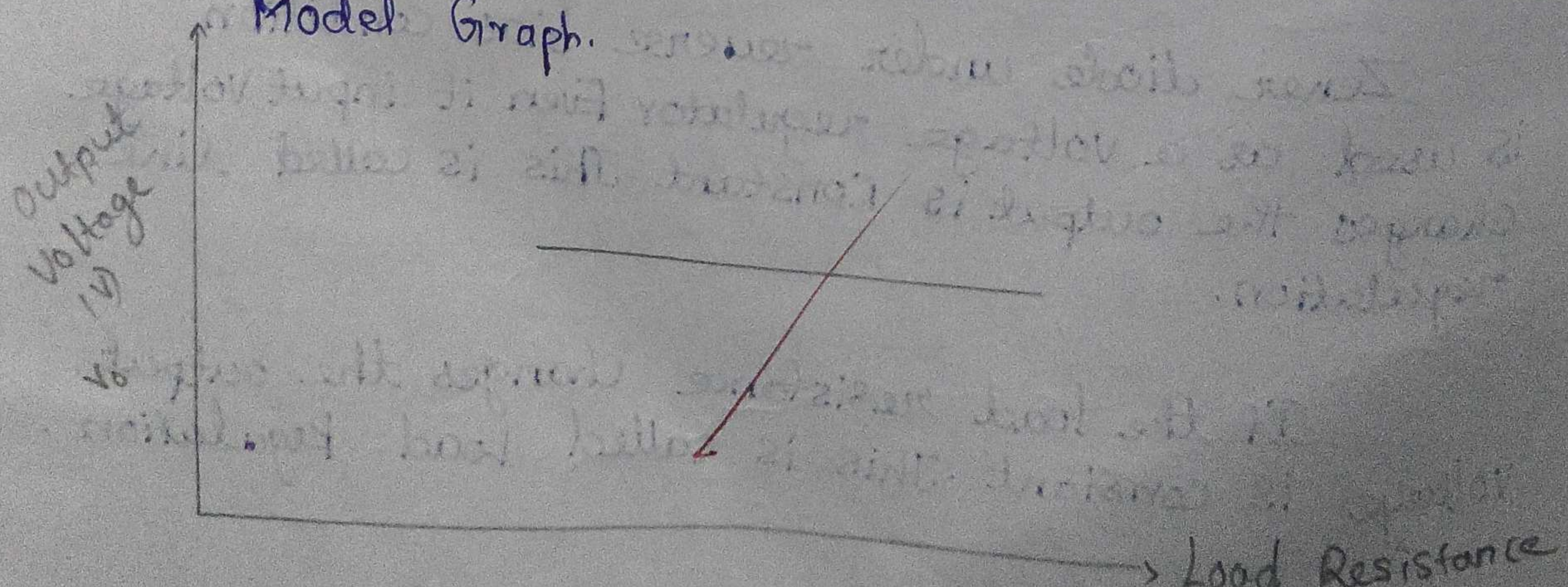
Circuit Diagram:



Tabulation:

Sl. No.	Load Resistance $R_L(\Omega)$	Output Voltage V_0 (V)
1	100 Ω	+5.19
2	200Ω	+5.20
3	300Ω	+5.20
4	400Ω	+5.20
5	500Ω	+5.20
6	600Ω	+5.21

Model Graph.



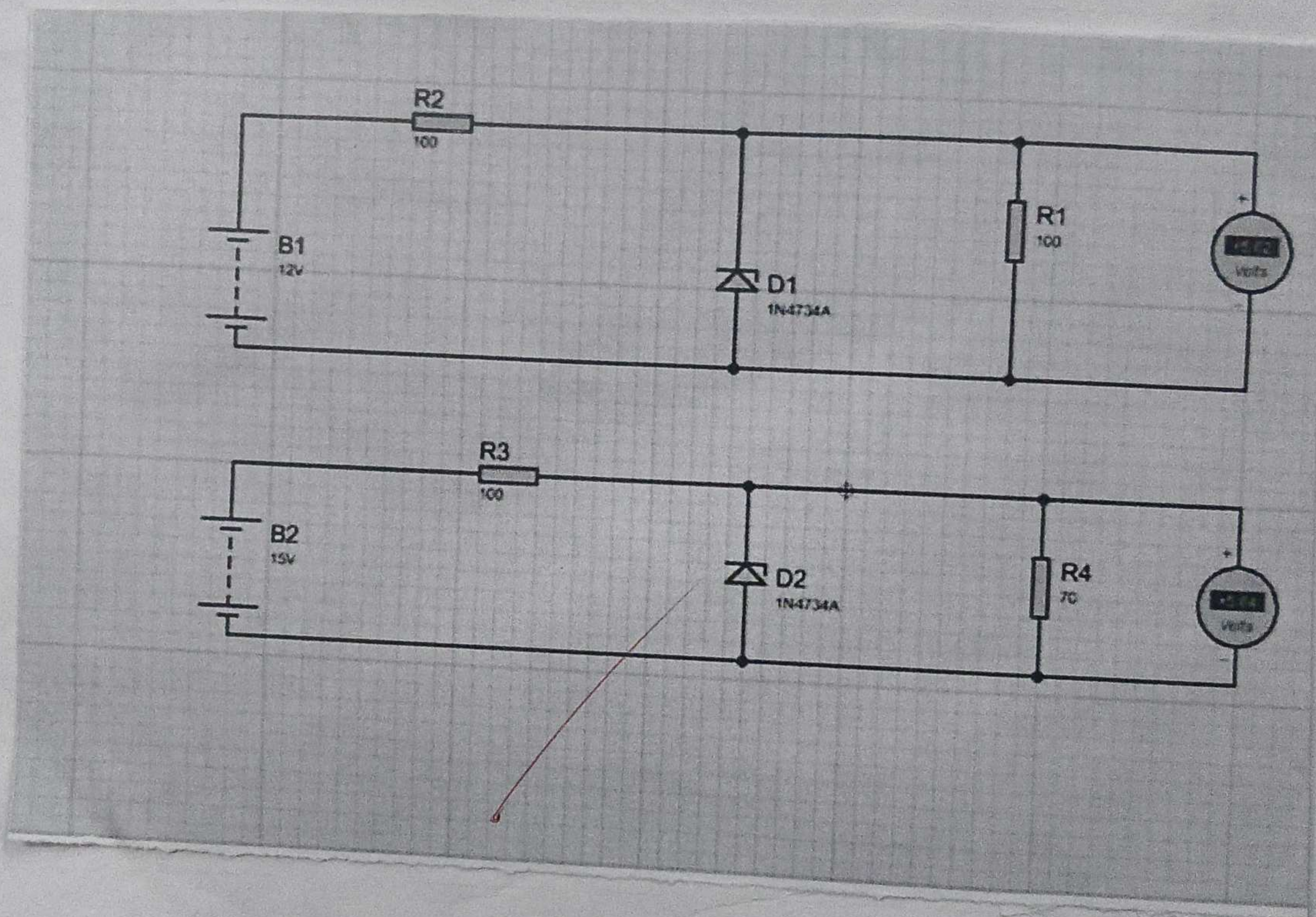
Procedure:

1. Drag the required components from the proteus library.

2. Connect the Components (required) as per Circuit diagram.

3. For line regulation vary the input Voltage and note the output Voltages in tabulation. Draw the graph between input and Output Voltages.

4. For load regulations vary the load resistance and note the output voltage in tabulation. Draw the graph between load resistance and output voltage.


R2
100
B1
12V
D1
1N4734A
R1
100
Volts
R3
100
B2
15V
D2
1N4734A
R4
70
Volts

Expt. No.

18/02/25

Result:

Thus Stimulation of line load regulation. operation of zener diode Success fully completed.